A sign-free expansion of double Grothendieck polynomials in double Schubert polynomials

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October 8, 2026

Abstract

We give a manifestly nonnegative combinatorial formula for the coefficients $a_{v,w}(\beta, z)$ in $\mathfrak{G}_v(x; z) = \sum_w a_{v,w}(\beta, z)\, \mathfrak{S}_w(x; z)$, the expansion of a double β-Grothendieck polynomial in the ordinary double Schubert basis, with no change of variables in either alphabet. The coefficient $a_{v,w}$ is the weight generating function of triples (pipe dream, WC graph with reduced co-pipe-dream, reduced pipe dream) glued along a Demazure product and a length-additive factorization; every triple contributes a single monomial in $\mathbb{N}[\beta, z]$. The formula composes three individually sign-free expansions: a Cauchy factorization of $\mathfrak{G}_v(x; z)$ over Demazure products, the integer expansion of a single Grothendieck polynomial in single Schubert polynomials, and the standard splitting of a single Schubert polynomial into double Schubert polynomials. It has been verified against direct computation for every $v \in S_n$, $n \leq 4$, and on samples in S_5. A second, independent conjecture on a staircase-dressed transition in the formal-inverse alphabet is recorded separately.

1 Conventions

Throughout, β and $x_1, x_2, \ldots, z_1, z_2, \ldots$ are independent formal variables and $\ell(w)$ is the inversion number. Permutation multiplication is composition, $(uv)(i) = u(v(i))$. Bruhat order means ordinary (strong) Bruhat order.

The multiplicative formal group law is $a \oplus b = a + b + \beta ab$. Double Schubert polynomials use the *difference* convention, $\mathfrak{S}_{s_1}(x; z) = x_1 - z_1$, and double Grothendieck polynomials the *plus* convention, $\mathfrak{G}_{s_1}(x; z) = x_1 \oplus z_1$. Concretely, with $\mathrm{RPD}(w)$ the reduced pipe dreams of w and $\mathrm{PD}(w)$ all (possibly unreduced) pipe dreams of w—i.e. cross sets whose row-reading word has Demazure product w—

$$\mathfrak{S}_w(x; z) = \sum_{R \in \mathrm{RPD}(w)} \prod_{(r,c) \in R} (x_r - z_c), \tag{1}$$

$$\mathfrak{G}_w(x; z) = \sum_{D \in \mathrm{PD}(w)} \beta^{|D| - \ell(w)} \prod_{(r,c) \in D} (x_r \oplus z_c). \tag{2}$$

A cross at (r, c) is the letter s_{r+c-1}; words are read row by row, right to left within a row. We write $\mathrm{exc}(D) = |D| - \ell(w)$ for the excess and $z^{\mathrm{wt}(D)} = \prod_{(r,c) \in D} z_r$ for the row monomial. In particular $\mathfrak{G}_v(x; z)|_{\beta=0} = \mathfrak{S}_v(x; -z)$, and the single polynomials are $\mathfrak{S}_w(x) = \mathfrak{S}_w(x; 0)$, $\mathfrak{G}_w(x) = \mathfrak{G}_w(x; 0)$.

For $u, v \in S_n$ we write $u \star v$ for the Demazure (0-Hecke) product: fold a reduced word of v onto u from the left, keeping each letter only when it increases length. In schubmult this is `Permutation.__matmul__`, i.e. `u @ v`.

2 The expansion

Fix $v \in S_n$. We seek

$$\mathfrak{G}_v(x;z) = \sum_{w \in S_n} a_{v,w}(\beta, z)\, \mathfrak{S}_w(x;z), \qquad a_{v,w} \in \mathbb{Z}[\beta, z]. \tag{3}$$

With $\deg \beta = -1$ and $\deg z_i = 1$, $\mathfrak{G}_v$ is homogeneous of degree $\ell(v)$ and $\mathfrak{S}_w$ of degree $\ell(w)$, so $a_{v,w}$ is homogeneous of degree $\ell(v) - \ell(w)$: $a_{v,w} = \sum_{k \geq 0} \beta^k p_k(z)$ with p_k homogeneous of degree $\ell(v) - \ell(w) + k$. In particular $a_{v,w}(\beta, 0) = 0$ unless $\ell(w) \geq \ell(v)$, while for $\ell(w) < \ell(v)$ the coefficient is a polynomial in z of degree $\ell(v) - \ell(w)$ at $\beta = 0$.

2.1 Three sign-free ingredients

Proposition 2.1 (Cauchy factorization over Demazure products)**.** *For every $v \in S_n$,*

$$\mathfrak{G}_v(x;z) = \sum_{\substack{u, v' \in S_n \\ u \star v' = v}} \beta^{\ell(u) + \ell(v') - \ell(v)}\, \mathfrak{G}_{u^{-1}}(z)\, \mathfrak{G}_{v'}(x). \tag{4}$$

This is the Cauchy-type identity of Fomin and Kirillov for double Grothendieck polynomials, transported to the present conventions; note the inverse on the z-factor. It is verified symbolically for all $v \in S_n$, $n \leq 4$, in `tests/mult/test_groth_double.py`. Each factor $\mathfrak{G}_{u^{-1}}(z)$ is, by (2) at second alphabet zero,

$$\mathfrak{G}_{u^{-1}}(z) = \sum_{P \in \mathrm{PD}(u^{-1})} \beta^{\mathrm{exc}(P)} z^{\mathrm{wt}(P)},$$

a sum of monomials with coefficient $+1$.

Proposition 2.2 (Single Grothendieck in single Schubert)**.** *For every $v' \in S_n$ there are nonnegative integers $c_{v',u'}$ with*

$$\mathfrak{G}_{v'}(x) = \sum_{u' \geq v'} \beta^{\ell(u') - \ell(v')}\, c_{v',u'}\, \mathfrak{S}_{u'}(x). \tag{5}$$

Nonnegativity of this expansion is due to Lenart. schubmult computes $c_{v',u'}$ by `WCGraph.groth_to_schub`: it is the number of pipe dreams $D \in \mathrm{PD}(v')$ in the $n \times n$ staircase whose *co-pipe-dream* $\mathrm{co}(D)$ is reduced with $\mathrm{Dem}(\mathrm{co}(D))\, w_0 = u'$. Here $\mathrm{co}(D)$ exchanges crosses and elbows and then reflects across the anti-diagonal, $(r, c) \mapsto (n + 1 - r - c, c)$. We take this realization as given; only the nonnegativity and integrality of $c_{v',u'}$ are used below.

Proposition 2.3 (Single to double Schubert)**.** *For every $u' \in S_n$,*

$$\mathfrak{S}_{u'}(x) = \sum_{\substack{u' = aw \\ \ell(a) + \ell(w) = \ell(u')}} \mathfrak{S}_a(z)\, \mathfrak{S}_w(x;z), \qquad \mathfrak{S}_a(z) = \sum_{R \in \mathrm{RPD}(a)} z^{\mathrm{wt}(R)}. \tag{6}$$

This is the $t = 0$ case of the standard identity $\mathfrak{S}_{u'}(x;t) = \sum_{u' = aw} \mathfrak{S}_a(z;t) \mathfrak{S}_w(x;z)$ over length-additive factorizations.

2.2 The formula

Substituting (5) and (6) into (4) and collecting the coefficient of $\mathfrak{S}_w(x;z)$:

Theorem 2.4. *For $v, w \in S_n$,*

$$\boxed{a_{v,w}(\beta,z) = \sum_{u\star v'=v} \sum_{u'\geq v'} \sum_{\substack{a:\ aw=u' \\ \ell(a)+\ell(w)=\ell(u')}} \beta^{\ell(u)+\ell(u')-\ell(v)}\ c_{v',u'}\ \mathfrak{G}_{u^{-1}}(z)\ \mathfrak{S}_a(z).} \tag{7}$$

Equivalently, $a_{v,w}$ is the weight generating function of triples (P, D, R) with

$$P \in \mathrm{PD}(u^{-1}), \qquad D \in \mathrm{PD}(v'),\ \mathrm{co}(D) \text{ reduced},\ \mathrm{Dem}(\mathrm{co}(D))w_0 = u', \qquad R \in \mathrm{RPD}(a),$$

subject to $u \star v' = v$ and $aw = u'$ with lengths adding, each triple weighted by the monomial

$$\beta^{|P|+\ell(u')-\ell(v)}\ z^{\mathrm{wt}(P)}\ z^{\mathrm{wt}(R)}.$$

In particular $a_{v,w} \in \mathbb{N}[\beta, z]$.

Proof. Immediate from Propositions 2.1–2.3; the β-exponent in (7) is $(\ell(u)+\ell(v')-\ell(v))+(\ell(u')-\ell(v'))$, and the pipe dream P contributes a further $\beta^{\mathrm{exc}(P)} = \beta^{|P|-\ell(u)}$ inside $\mathfrak{G}_{u^{-1}}(z)$, for a total of $|P| + \ell(u') - \ell(v)$. Each ingredient is a sum of monomials with coefficient $+1$, and the weight of a triple is the product of the three weights. □

Remark 2.5. No variable is substituted anywhere: both alphabets x, z are the original ones, the output basis is $\{\mathfrak{S}_w(x;z)\}$, and the coefficients involve only β and z. The sum over (u, v') runs over the Bruhat interval below v (both $u \leq v$ and $v' \leq v$ when $u \star v' = v$); the sum over u' runs over the interval above v'.

Remark 2.6. The degree bookkeeping of (3) is visible in (7): $\mathfrak{G}_{u^{-1}}(z)\mathfrak{S}_a(z)$ has degree $\ell(u) + \ell(a)$ and the β-power contributes $-(\ell(u) + \ell(a) + \ell(w) - \ell(v))$, for a total of $\ell(v) - \ell(w)$.

2.3 Examples

Example 2.7 ($n = 2$)**.** Pairs with $u \star v' = s_1$ are $(e, s_1), (s_1, e), (s_1, s_1)$. Using $\mathfrak{G}_{s_1}(z) = z_1$ and $c_{s_1,s_1} = 1$, $\mathfrak{S}_{s_1}(x) = \mathfrak{S}_{s_1}(x;z) + z_1$:

$$\mathfrak{G}_{21}(x;z) = (1+\beta z_1)\,\mathfrak{S}_{21}(x;z) + (2z_1 + \beta z_1^2)\,\mathfrak{S}_e(x;z).$$

Note $2z_1 + \beta z_1^2 = z_1 \oplus z_1$.

Example 2.8 ($v = 132$)**.** Writing $A_i = 1 + \beta z_i$, the nonzero coefficients are

w	$a_{132,w}$
132	$A_1^2 A_2$
231	$\beta A_1 A_2$
e	$(A_1^2 A_2^2 - 1)/\beta \ = \ z_1 \oplus z_1 \oplus z_2 \oplus z_2$

There is no $\mathfrak{S}_{21}$ term. The e-coefficient is the polynomial $2z_1 + 2z_2 + \beta(z_1^2 + 4z_1z_2 + z_2^2) + 2\beta^2(z_1^2 z_2 + z_1 z_2^2) + \beta^3 z_1^2 z_2^2$, all of whose coefficients are positive, as the theorem predicts.

2.4 Verification and implementation

The formula is implemented as `schubmult.mult.groth_double.dgroth_to_dschub_positive`. It enumerates $\mathrm{PD}(u^{-1})$ via `WCGraph.all_wc_graphs`, $c_{v',u'}$ via `WCGraph.groth_to_schub`, and $\mathrm{RPD}(a)$ via `RCGraph.all_rc_graphs`, multiplying the resulting monomials; no algebraic expansion routine is called. It agrees with the exact (divided difference) expansion `dgroth_to_dschub` for every $v \in S_n$, $n \le 4$, and on samples in S_5; the coefficients are checked to involve only β, z and to have positive integer monomial coefficients. The Cauchy factorization (4) is tested separately. The routine enumerates two Bruhat intervals and is a research implementation rather than a multiplication kernel.

3 A staircase-dressed transition (conjectural)

A different, earlier line of computation produced an unsigned graph formula in a *dressed* basis. We record it because it is verified much further (S_6) than its ordinary counterpart would be, and because its objects—reduced dreams whose complements are unreduced—are of independent combinatorial interest. It is *not* needed for Theorem 2.4.

Let $N = \binom{n}{2}$, $w_0 \in S_n$ the longest element, $\bar{x}_i = -x_i/(1+\beta x_i)$ the formal inverse, $A_c = 1+\beta z_c$, and $P_n(x) = \prod_{i=1}^{n-1}(1+\beta x_i)^{n-i}$. Define coefficients $C^{(n)}_{v,\sigma}$ by

$$\mathfrak{G}_v(x;z) = (-1)^{\ell(v)}\, P_n(x) \sum_{\sigma \in S_n} C^{(n)}_{v,\sigma}(\beta, z)\, \mathfrak{S}_\sigma(\bar{x}; z). \tag{8}$$

This is the fixed-staircase case of the "phantom expansion" used in schubmult's Grothendieck v-path kernel, which computes $C^{(n)}_{v,\sigma}$ by a signed divided-difference recurrence on states (permutation, multiset of surviving atoms A_c).

Conjecture 3.1. *For $v, \sigma \in S_n$,*

$$C^{(n)}_{v,\sigma} = \beta^{\ell(\sigma)-\ell(v)} \sum_{\substack{D \in \mathrm{RPD}(\sigma w_0) \\ v \le \mathrm{Dem}(\mathrm{co}(D))}} \prod_{(r,c)\in D} A_c,$$

the sum over reduced pipe dreams D for σw_0 in the full $n \times n$ staircase whose co-pipe-dream has Demazure permutation at least v in Bruhat order. The co-pipe-dream may be unreduced.

Since D is reduced, $|D| = N - \ell(\sigma)$, so each term is homogeneous of degree $N - \ell(\sigma)$ in the atoms, and $v \le \mathrm{Dem}(\mathrm{co}(D))$ forces $\ell(\sigma) \ge \ell(v)$. The conjecture matches the recurrence—not just the coefficients but each individual atom multiset with its integer multiplicity—for all 720 inputs in S_6 (6080 reduced dreams), and the identity (8) is checked symbolically through S_3. It is implemented as `dgroth_copipe_expansion`.

Two features distinguish it from the integer expansion (5): the acceptance condition is an upper Bruhat filter rather than an exact permutation, and the complement is allowed to be unreduced. For instance, with $n = 3$, $v = 132$, $\sigma = 231$: the reduced dream $\{(2,1)\}$ for $\sigma w_0 = 132$ has complement with word $s_2 s_2$, Demazure permutation $132 \ge v$; it is accepted and contributes βA_1, which is needed.

Remark 3.2. Attempts to turn (8) into the ordinary expansion (3) by alphabet manipulation produced coefficients that were either signed (expanding a first-alphabet prefactor in Schubert polynomials) or positive but rational (changing the coefficient alphabet $\bar{z} \to z$, with local weights $z_r + z_c/A_c$ whose denominators cancel only after aggregation). Theorem 2.4 avoids both by never leaving the original alphabets.

4 Open problems

1. **A single-object formula.** Theorem 2.4 sums over *triples* of graphs. The $z = 0$ specialization collapses to the single family of (5). Is there a single family of decorated pipe dreams for v, with a local weight in β, z, whose generating function is $a_{v,w}$? The factorization $x_r \oplus z_c = A_c(x_r - z_c) + (z_c \oplus z_c)$ applied cell by cell to (2) suggests marking cells of a pipe dream of v, but the unmarked product $\prod(x_r - z_c)$ over a non-reduced sub-dream is not a Schubert polynomial, which is precisely the obstruction that the Cauchy step circumvents.

2. **Proof of Conjecture 3.1**, and its relation to Theorem 2.4: the conjecture's objects are reduced dreams of σw_0 with a Bruhat filter on the complement, the theorem's are unreduced dreams of v' with an exact complement; a bijective explanation of how the Bruhat filter and the Cauchy sum over $u \star v' = v$ trade off would illuminate both.